

DEPI - 2

Digital Arts - Mastering Professional 3D Designs

2024-2025

Project category



Product design
Perfume

Product design

DEPI-2

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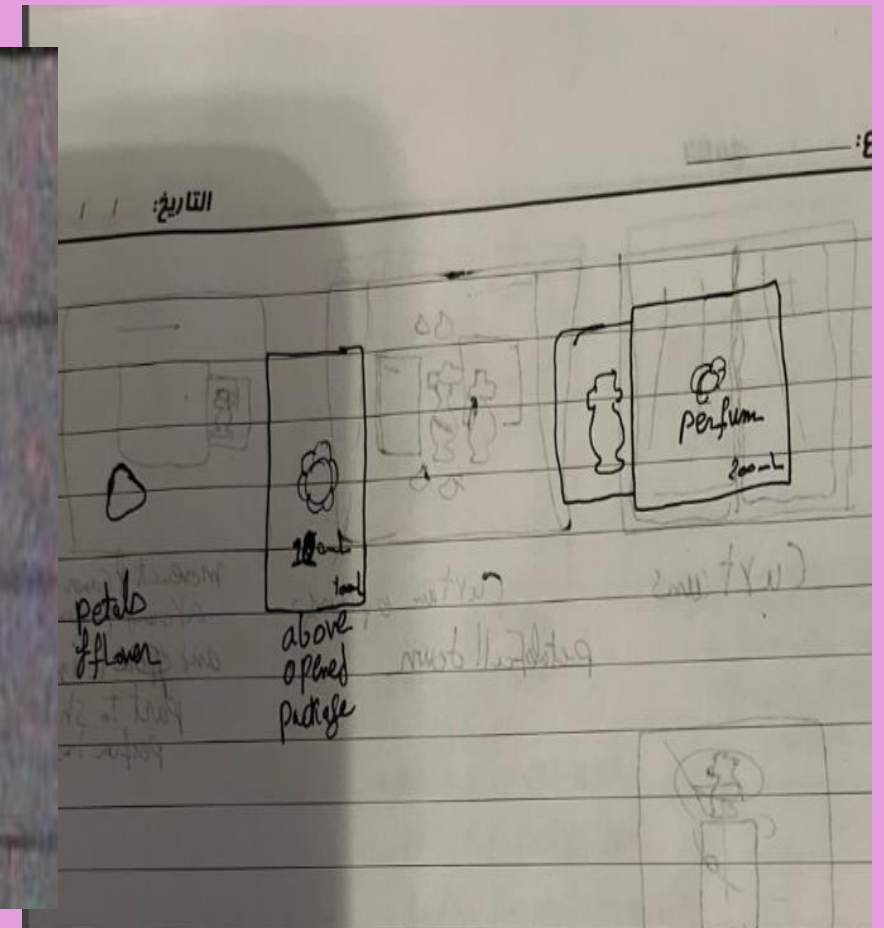
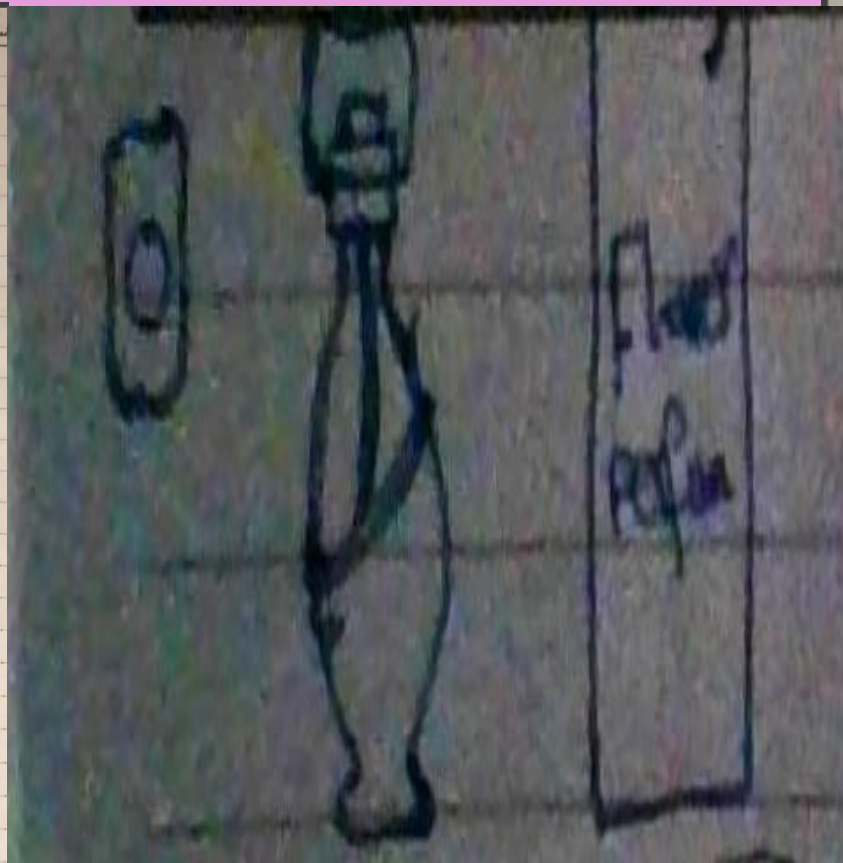
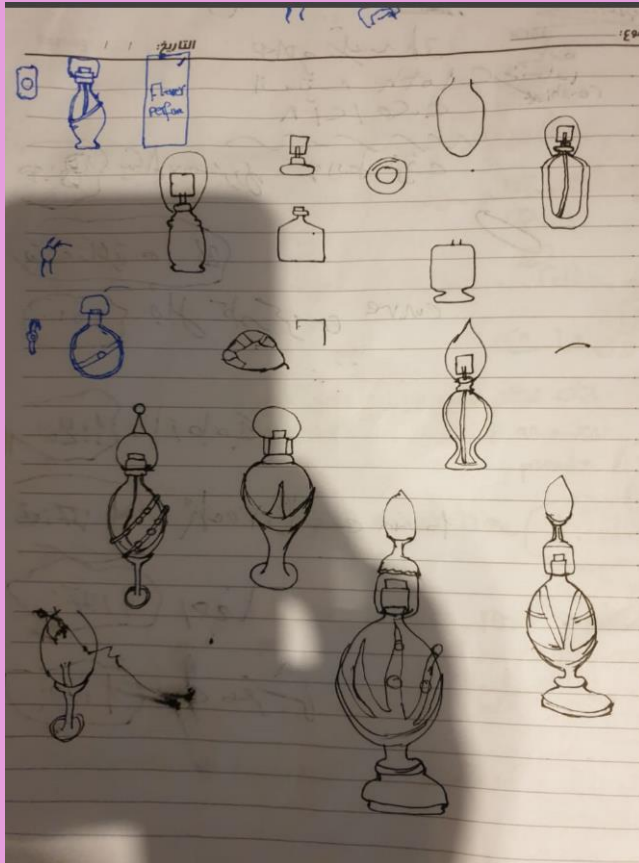
Stages of project :

1. Pre-production
2. Production
3. Post-production



1-Pre-Production:

Product design sketching and ideas for perfume and its packages .



1-Pre-Production:

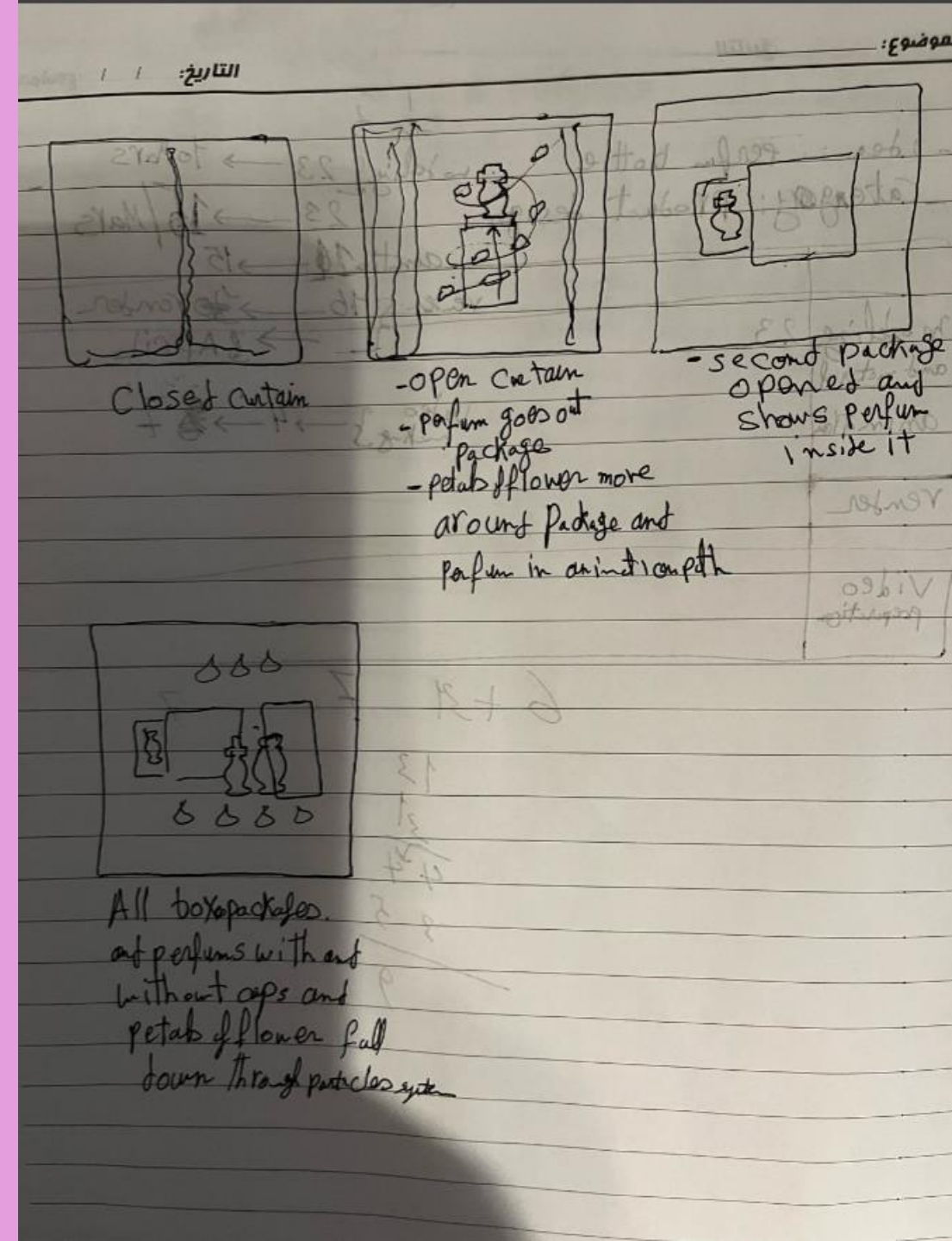
Choose perfume bottle design



1-Pre-Production:

Making storyboard for animation of four scenes.

- Imagine how to show and animate the product in four scenes.



2-Production stage:

- **Scene Organization and Optimization:**
 - Initially, all scenes were placed within a single 3ds Max file.
 - Upon advice from our the supervisor and trainer, the project was split into four separate files, improving workflow efficiency, especially in lighting and rendering time .

2-Production stage:

- **Tools:**

- 3D Software for modeling : 3d Max 2025

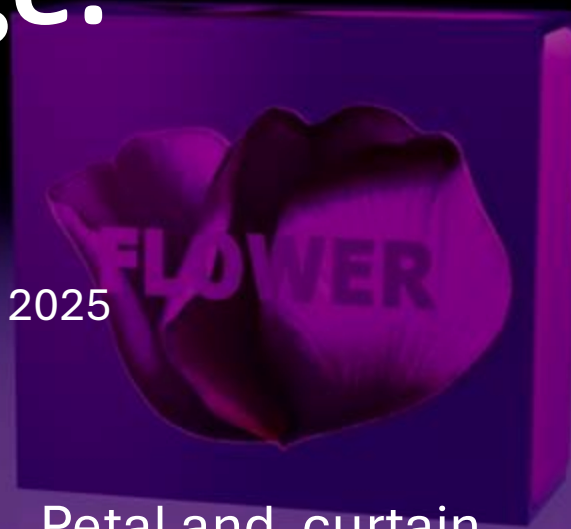
- **Models and material :**

- Perfume bottle ,Two package ,Petal and curtain

- **Lighting :** Studio light

- **Camera:** Physical camera

- **Renderer:** Arnold

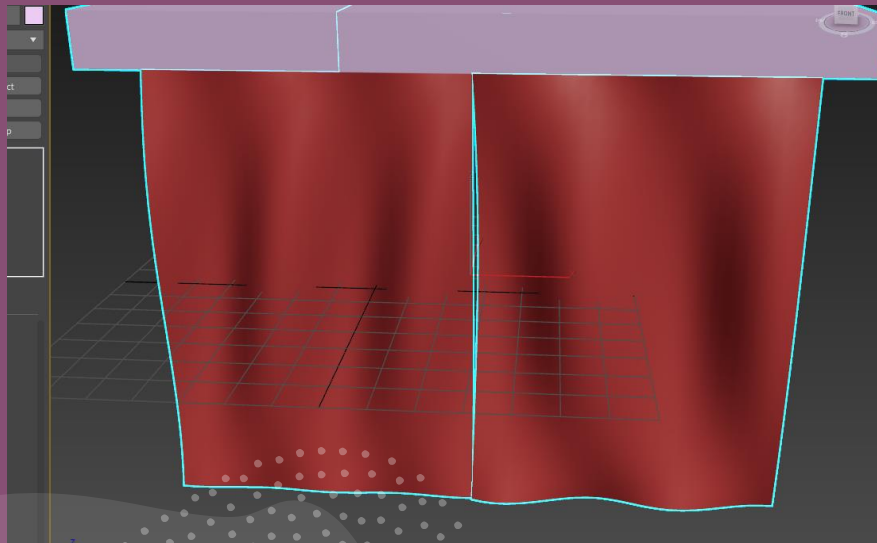


2-Production stage:

- **Useful Modifiers and Solutions to Challenges:**
- **Noise Modifier:** Added complexity and realism to the curtain.
- **Cloth modifier:** Simulating the motion of cloth as it interacts with the environment.
- **TurboSmooth Modifier:** Smoothed models, increased polygons, and eliminated sharp edges.
- **Arnold Properties:**
 - Adjusted model interaction with specific lights.
 - Enabled light inclusion/exclusion for enhanced visual control.
- **Edit Poly:**
 - Enhanced object control at sub-object levels (vertex, edge, polygon, and element).

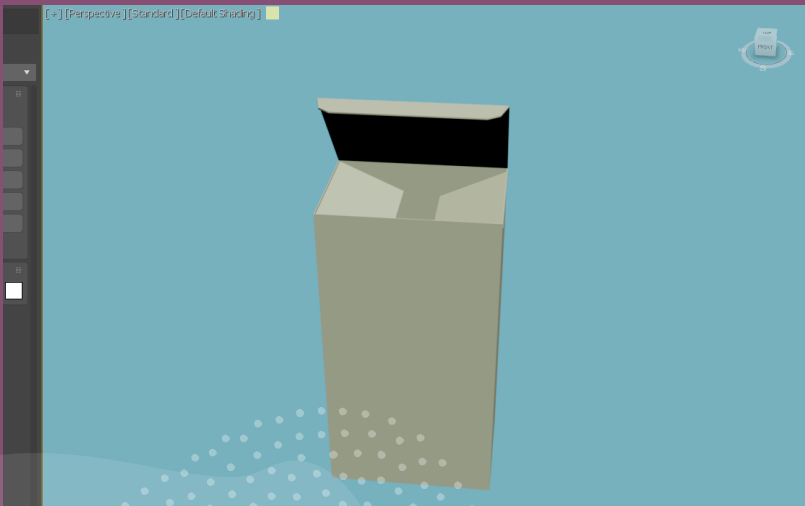
2-Production stage:

- Curtain model and material :



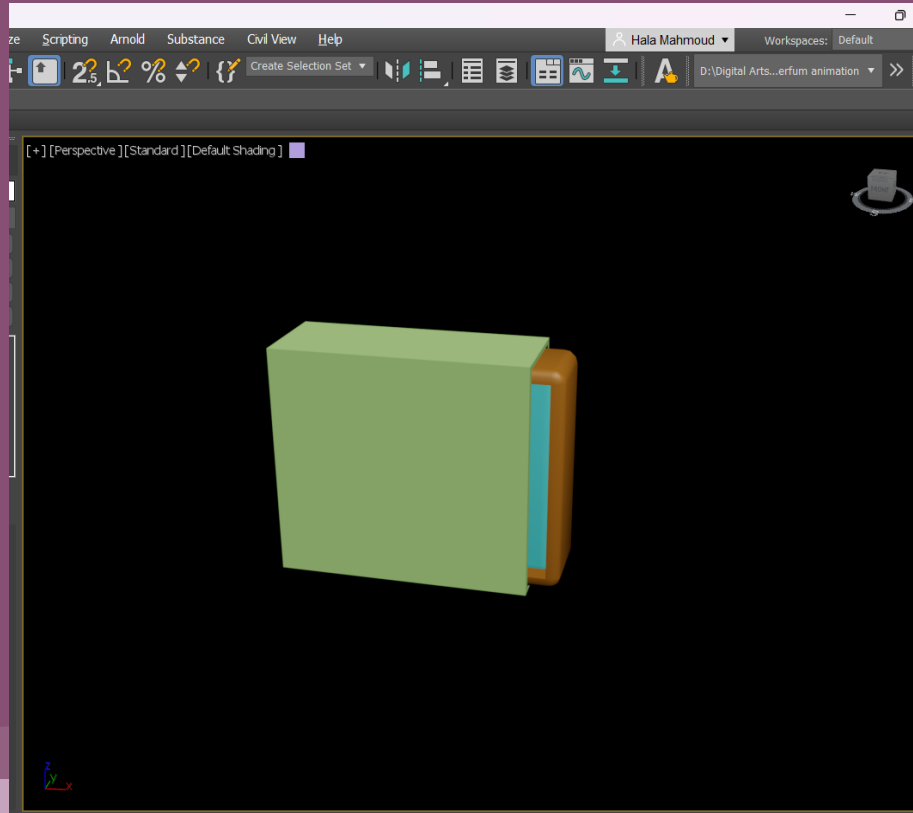
2-Production stage:

- package models and material:



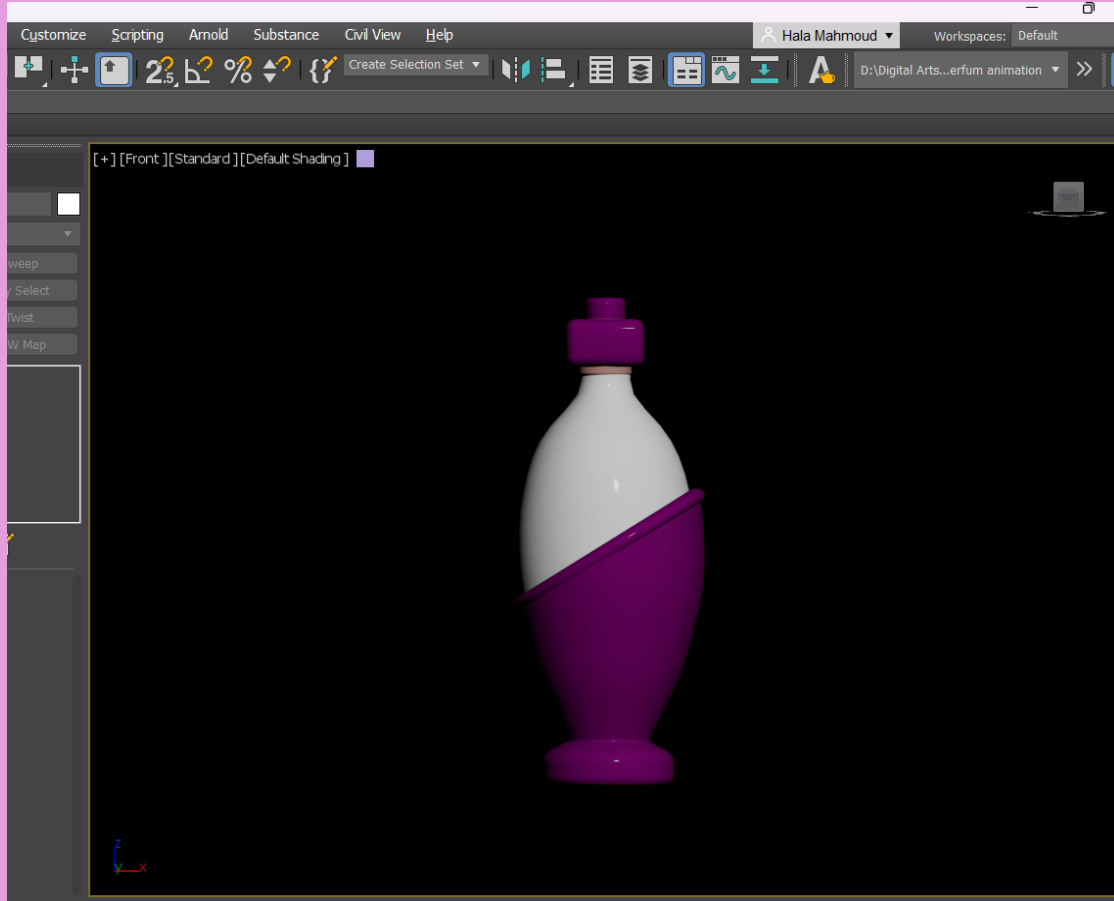
2-Production stage:

- package models and material:



2-Production stage:

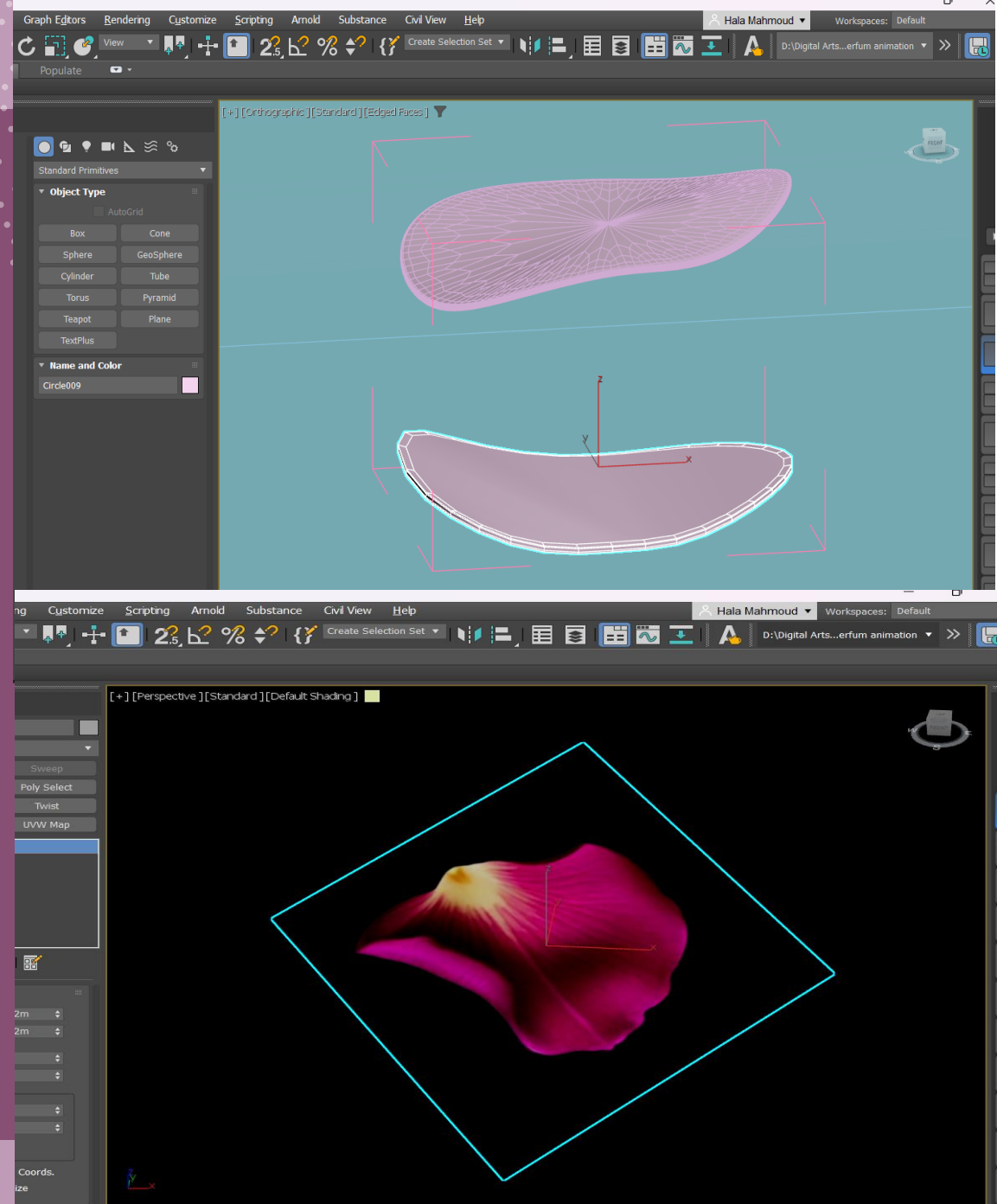
- Perfume models and material:



2-Production stage:

Petal model and material:

- Create petal model was used in first scene
- Main petals were created by plane with petal image on Bitmap lookup to decrease render time



2-Production stage:

Lighting and Background Setup

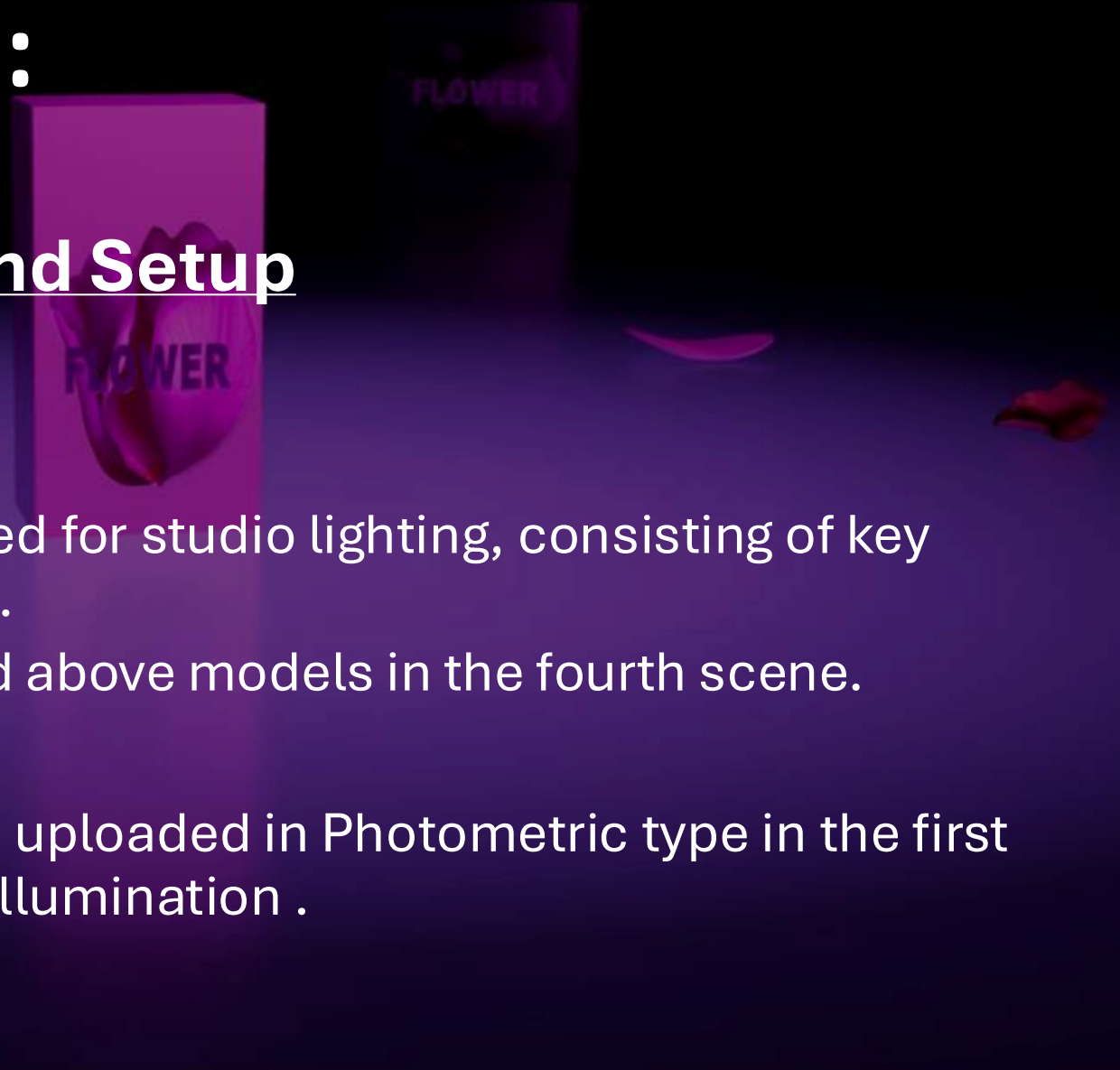
- **Lighting setup:**

- Arnold Light Types:**

- Quad (Plane Light) was used for studio lighting, consisting of key light, fill light, and rim light.
 - Additional light was placed above models in the fourth scene.

- Photometric Light:**

- A light file(IES profile) was uploaded in Photometric type in the first scene to achieve realistic illumination .



2-Production stage:



- Background Setup:

- A plane was converted into an editable poly.
- Edge modifications included chamfering the middle edge, increasing segments, and smoothing for curvature.

2-Production stage:

- **Animation:**
- **Keyframe Animation:** Applied to all four scenes directly on the timeline.
- **Total Keyframes:** 1,299 frames across four scenes
- **Time Configuration settings:**
 - Frame rate: 30 FPS
 - Format: NTSC
- **Path Constraint Animation:(in scene 2)**
 - Used to animate petals swirling around the perfume bottle in Scene 2.
 - A helix spline was utilized as the animation path.

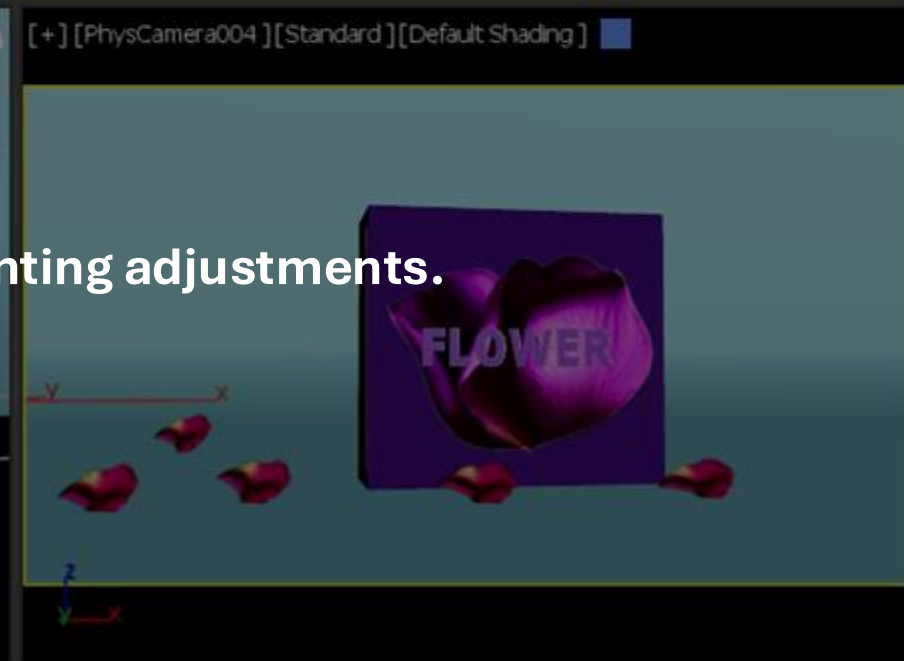
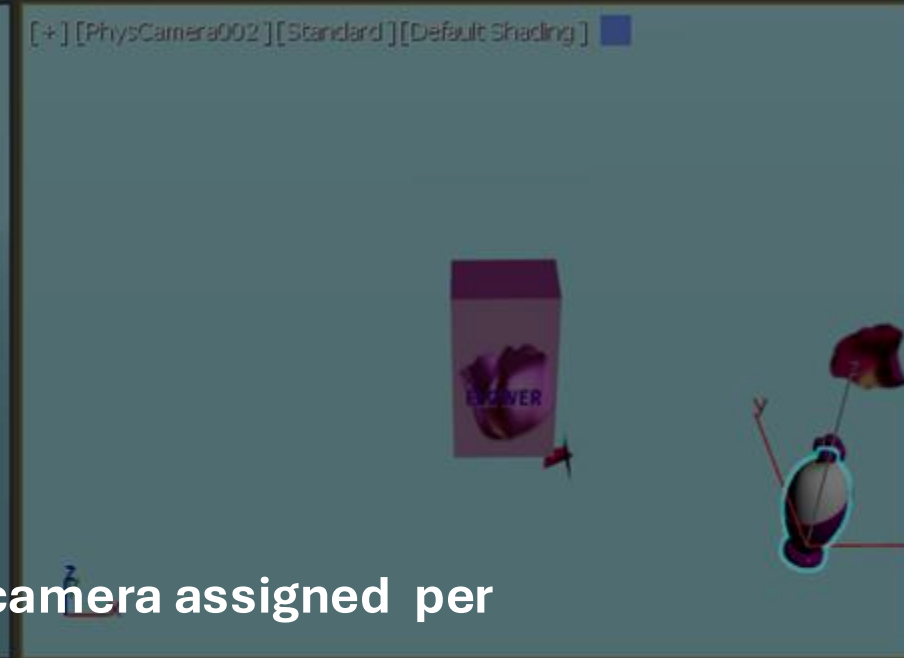
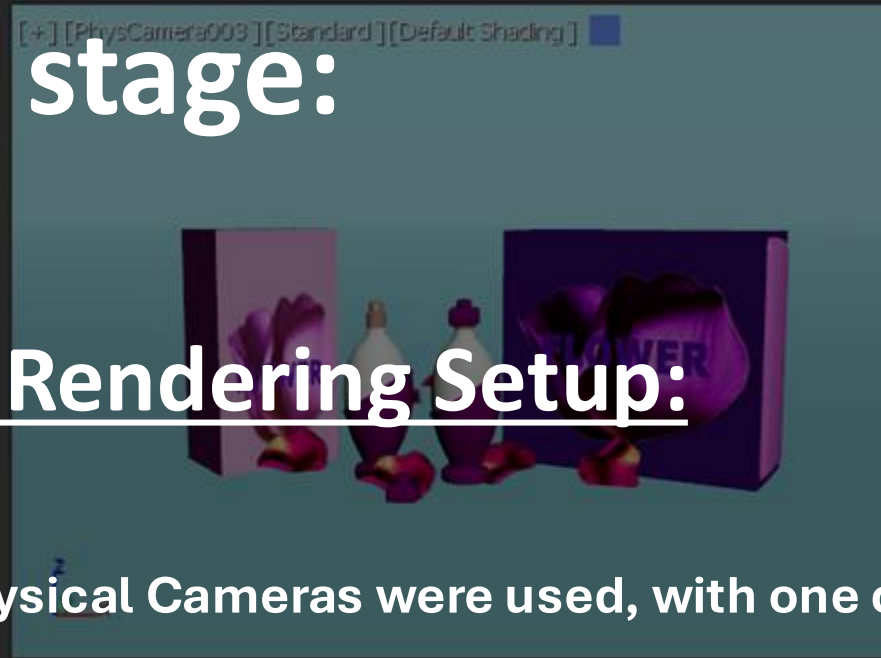
2-Production stage:

- **Curtain Animation: (in scene 1)**
 - Cloth modifier applied, simulated, and converted into keyframes.
- **Particle System: (in scene 4)**
 - **PF Source** was used to make petals fall from top to bottom.
 - **POmniFlect Deflector** in space warps enabled petals to land realistically on the ground.

2-Production stage:

Camera and Rendering Setup:

- **Camera:**
 - Four standard Physical Cameras were used, with one camera assigned per scene.
- **Rendering:**
 - Renderer: Arnold
 - Testing Mode: ActiveShade for testing material and lighting adjustments.
 - Final Render Mode: Production Rendering
 - Resolution: 1920x1080



3-Post-Production:

- After I got render output for four scenes ,I used blender software as video editor to convert images into video for resolution 1920x1020.
- Video duration : 1 minute and 15 seconds .
- Add music for video that is owned by FesliyanStudios(Please DO NOT add this audio content to the Youtube Content ID System. I have used background music which is owned by FesliyanStudios).
- Links of project folder :

<https://drive.google.com/drive/folders/18whbIPtGyKf84e7hxLkdwa-WQ4h-mXxv?usp=sharing>

- Link of video:

https://drive.google.com/drive/folders/1fuqliVVU2t7p4bJSXl0PAMwR_4si2s_7?usp=sharing

Acknowledgments

I would like to thank my supervisor and trainer,

Engineer Ahmad Tawfiq

For his knowledge and efforts with us

Thanks

